

With a leading focus of our clinic on longevity medicine, disease prevention, and brain regeneration, we have chosen to adopt the KetoFLEX 12/3 diet as our foundational dietary plan for the brain. The KetoFLEX diet was developed by Dr. Dale Bredesen and is part of the Bredesen Protocol for the treatment and prevention of Alzheimer's and vascular dementia. The power of the KetoFLEX diet is its flexibility in meeting the diversity of patients' abilities and health needs. The KetoFLEX is a modified keto diet designed to aid in preventing and reversing cognitive decline, and in addition, it reduces the risk of the leading diseases of aging and premature death.

Through his research and a successful clinical trial, Dr. Bredesen has found the most effective nutrition for support of brain health includes a mix of the Mediterranean diet, a moderate ketogenic diet, and nightly fasting of a minimum of 12 hours, starting 3 hours before bed. This dietary approach is heavily plant-based, high in healthy fats, and nutrient-dense whole foods (emphasizing local, organic, seasonal, and colorful non-starchy vegetables), combined with an adequate amount of protein. This dietary approach abstains largely from refined sugar, grains, refined carbohydrates, processed foods, chemical and food additives, and dairy.

KetoFLEX 12/3 PLUS Overview

KetoFLEX 12/3 is a friendly lifestyle diet made up of flexible guides to optimize cognition, longevity, mood, and energy. This dietary approach is flexible and effective enough to also have significant benefits in diabetic patients, and reduce cardiovascular risk, and most inflammatory conditions.

Keto

- Keto refers to ketosis, the natural process by which a body breaks down stored and dietary fat to use as fuel in the absence of sugars and carbohydrates from starchy foods and grains. You are in a state of Ketosis starting approximately 4 hours after eating, with fasting, and when adhering to a strict Keto diet. The key to staying in Ketosis is avoiding foods that elevate insulin such as sugars, refined carbohydrates, starches, breads, pasta, cereals, and all grains. Fat usually has no effect on raising insulin. Generally, protein has minimal or no effect on insulin.

FLEX

- FLEX refers to metabolic flexibility, a flexitarian diet, and flexible grain intake.
- Metabolic flexibility is defined as the ability to use either glucose or fat as a fuel source. The longer we stay in ketosis, or in other words, the longer we exercise, work, and sleep in ketosis, our body becomes more able to convert to fat burning as a source of energy instead of waiting to be fed carbohydrates every 4-6 hours.
- FLEX also refers to a flexitarian diet, in which meat is optional. Focusing on generous healthy fats, vegetables, and plant-based proteins is also a viable option.
- FLEX also allows grains or starches at dinner for those who maximize Ketosis for the majority of the day by avoiding grains at breakfast and lunch, a very effective approach for diabetic or pre-diabetic patients or those with elevated insulin who desire to lose weight. The goal is to limit the consumption of grains to <50 grams daily. Keeping the consumption of the grains at one meal, maintains ketosis, fat burning, and promotes metabolic flexibility while providing more options and flexibility for family dinners.

12/3

- 12/3 refers to the minimum amount of time each day that you should spend fasting; for at least 12 hours and no food at least 3 hours before bed.
- With all the established benefits of extended fasting known as intermittent fasting, we encourage fasting up to 16 hours daily after the age of 40, limiting the eating window to 8 hours total daily instead of 12 as recommended in the KetoFLEX diet.

- It is our opinion that most individuals after age 40 could benefit from eating one less meal a day. This change could have significant benefits for diabetic or pre-diabetic patients and for those wishing to lose or maintain weight as they age.

PLUS

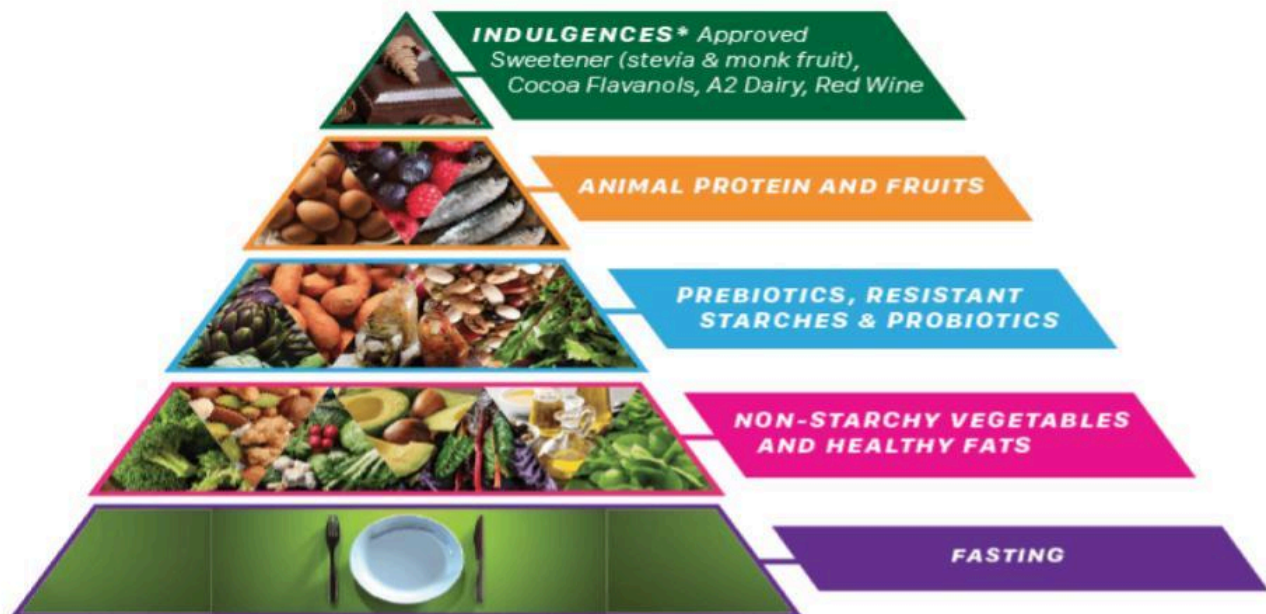
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Eliminate the Worst Inflammatory Foods

The greatest impact of the KetoFLEX 12/3 diet is not just what we are eating, but what we are avoiding while focused on good nutrition. It is just as important to know what NOT to do, as it is to know what to do. First eliminate the most common inflammatory foods pictured below including processed refined sugar, simple carbohydrates (breads, muffins, cakes, pastries, crackers, chips, pretzels, pasta, etc.), conventional dairy (milk, cheese, cottage cheese, yogurt), and grains (wheat, corn, rye, oats, rice, etc.). These foods are the leading contributors to insulin resistance, candida overgrowth, weight gain, afternoon fatigue, brain fog, and many other common symptoms mistakenly attributed to normal aging.

KetoFLEX 12/3 PLUS Food Pyramid

Below you will find a description of the most correct food pyramid according to research and not anything like the food pyramid of “olden” days designed by the FDA and special interest groups.



Level One: Fasting to promote cell regeneration through metabolic flexibility and autophagy

- Everyone should stop eating at least three hours before bed and extend the fast for at least twelve hours. Those who are ApoE4 positive can work on slowly extending the fast to 16+ hours.

- To learn about the benefits of fasting, including autophagy, often referred to as cellular regeneration, read Foundations for Fasting (Intermittent and Multi-Day Fasting).
- Get coaching for intermittent and multi day fasting to promote autophagy
- Add Ketones supplements and electrolytes in place of meal while fasting to reduce fatigue, brain, fog, hypoglycemia, irritability, and headaches

Level Two: Foods which should be freely indulged

This is the level of the pyramid that comprises the foods, non-starchy vegetables and healthy fats, that should make up the majority of your plate. See tips below for making smart choices.

Non-starchy Vegetables

- Non-starchy vegetables are those with a low glycemic index that won't spike your blood glucose or affect your ability to get into ketosis. Freely enjoy deeply pigmented, organic, seasonal, preferably local, non-starchy vegetables, including leafy greens and cruciferous, (some raw and some cooked) with fresh herbs, spices, and teas.

Healthy Fats

- Increasing healthy fats, such as high polyphenol extra virgin olive oil (EVOO), avocado, nuts, and seeds will not only make the nutrients from your vegetables more bioavailable, but it will also help you achieve nutritional ketosis.
- Coconut oil is full of healthy fatty acids to help induce ketosis, including keto-friendly medium-chain triglycerides (MCTs).
- Wild-caught seafood rich in healthy fats like salmon, sardines, and mackerel
- Grass-fed meat, in moderation

Level Three: Promote a healthy microbiome

The KetoFLEX diet also resolves the underlying chronic intestinal inflammation and promotes ongoing gut health by promoting a balanced and diverse microbiome. The gut microbiome provides the foundation for healthy function of our nutritional, immune, hormonal and neurological systems. The following foods promote healthy probiotic balance and growth and include prebiotic fibers, probiotic foods, and digestive resistant starches.

- Specific foods, including prebiotic fiber (artichoke hearts, jicama, dandelion greens, mushrooms, leeks, green banana, etc.), resistant starches (tubers, legumes, parsnips, cassava root, rutabagas, pistachios, etc.) and probiotic foods (fermented vegetables, sauerkraut, kimchi, low sugar kombucha, tempeh, miso, etc.) in small quantities as tolerated can help to cultivate a healthy gut.

Level Four: To be consumed in moderation and wisely

This level of the pyramid includes animal protein and fruit, both of which can be very healthful, but should be chosen carefully and intake moderated. See tips below.

Animal Protein

- It's very important to get an adequate amount of protein but consuming too much impedes autophagy (also fostered by fasting), metabolic health, and even longevity. Healthy people, under age 65, should limit protein consumption to 0.8-1.0 gram per kilogram of lean body mass per day. Concurrently implementing a strength training program to avoid frailty is highly recommended.

- When consuming animal protein, quality matters. We recommend prioritizing low mercury, wild-caught seafood for the DHA and eggs from pastured hens for the choline. When eating meat, be sure that it's 100% pastured.

Fruit

- Ancestrally, fruit was enjoyed seasonally to fatten up for the winter. Because fruit is now imported from around the globe to be available year round, many people over consume and choose varieties that are hybridized to be sweeter and lower in fiber, impairing metabolic health. Fruit can be very nutritious when carefully selected.
- Eat heirloom fruit seasonally. Always balance nutritive value against glycemic concerns. Exceptions include small portions of neuroprotective wild berries, lemons, limes, and unripened tropical fruit (green plantains, bananas, mangoes, papayas, and kiwi) as resistant starch and for their natural digestive enzymes

Level Five: To be used sparingly

Just as the descriptor indulgences suggests, this is the level of the pyramid where we really start to put on the brakes. While all of the foods included on this level, grains, approved sweeteners, cocoa flavanols, A2 dairy, and dry red wine, can be quite healthful for some, they all need to be enjoyed in moderation.

Grains and refined carbs

- EVND allows minimal grains or starches at dinner for those who maximize Ketosis for the majority of the day by avoiding grains at breakfast and lunch. The goal is to limit consumption of grains to <50 grams daily. Keeping the consumption of the grains at one meal, maintains ketosis, fat burning, and promotes metabolic flexibility during the majority of the day, while providing more options and flexibility for at least one meal daily.

Approved Sweeteners

- If necessary, use limited amounts of approved sweeteners that include pure versions of stevia and monk fruit. Local honey may be used by those who are insulin sensitive.

Cocoa Flavanols

- To derive the health benefits of flavanols, enjoy small amounts of high cacao chocolate, low in sugar, cadmium and lead.

A2 Dairy (Less Inflammatory)

- You may have small amounts of A2 dairy (such as goat cheese) if tolerated and desired.

Dry Red Wine

- Alcohol is a neurotoxin and best avoided by anyone with dependence issues or at risk of or suffering with cognitive decline. If you decide to occasionally indulge, consider small amounts of organic sugar-free, low alcohol red wine.